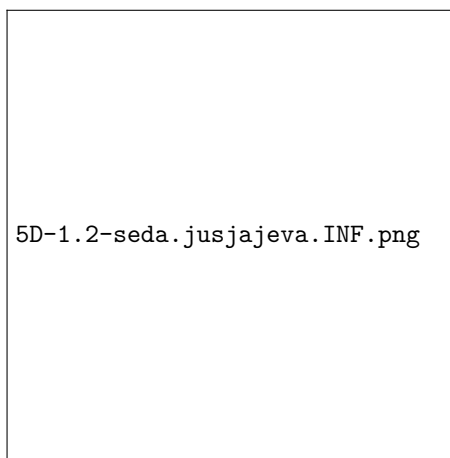


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Studierichting: Informatica
Oefeningenreeks 5D nr. 1.2

$$\int_{-2}^1 (x^3 - x) dx$$
$$\int (x^3 - x) dx = \int x^3 dx - \int x dx = \frac{x^4}{4} - \frac{x^2}{2} + c = \frac{x^4 - 2x^2}{4} + c$$

Grafiek:



$$\begin{aligned} \Rightarrow \int_{-2}^1 (x^3 - x) dx &= - \int_{-2}^{-1} (x^3 - x) dx + \int_{-1}^0 (x^3 - x) dx - \int_0^1 (x^3 - x) dx \\ &= - \left[\frac{x^4 - 2x^2}{4} \right]_{-2}^{-1} + \left[\frac{x^4 - 2x^2}{4} \right]_{-1}^0 - \left[\frac{x^4 - 2x^2}{4} \right]_0^1 \\ &= - \left(\frac{1-2}{4} - \frac{2^4-2(2)^2}{4} \right) + \left(0 - \frac{1-2}{4} \right) - \left(\frac{1-2}{4} - 0 \right) \\ &= \frac{9}{4} - \frac{1}{4} + \frac{1}{4} \\ &= \frac{11}{4} \end{aligned}$$

References